Topology (2025 Fall)

1. (20 points) Consider the finite-complement topology on $\mathbb{R}$, i.e. $A \subset \mathbb{R}$ is closed if and only if A is finite or $A = \mathbb{R}$. Prove that every subspace of $\mathbb{R}$ is compact.
2. (15 points) Let (X, d) be a compact metric space. There exists $0 < \alpha < 1$ such that

   $$d(f(x), f(y)) \leq \alpha d(x, y), \forall x, y \in X$$

   Prove that f has unique fixed point.
3. (20 points) A topological space X is called *locally path-connected* if for any $x \in X$ and neighborhood U of x, there exists neighborhood V of x and $V \subset U$ such that V is path-connected. Prove that if X is connected and locally path-connected, then X is path-connected.
4. (15 points) Let $\mathrm{GL}(n, \mathbb{C}) = \{A \in \mathrm{Mat}_{n\times n}(\mathbb{C}) \mid \det A \neq 0\}$. Prove that $\mathrm{GL}(n, \mathbb{C})$ is path-connected.
5. (15 points) Let X be a topological space, there exists a deformation retraction from X to $\{p\} \subset X$. For any neighborhood U of p, prove that there exists a neighborhood V of p such that the inclusion map $i : V \to U$ is homotopic to a constant map.
6. (15 points) Compute the fundamental group of $S^2 \setminus \{p_1, p_2\}$, where $p_1 \neq p_2 \in S^2$.